NATURAL ENEMY COMPLEX OF COCONUT LEAF EATING CATERPILLAR

Opisina arenosella Walker (Lepidoptera: Xylorictidae) in Karnataka

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Natural enemy complex of coconut leaf eating caterpinar~ Opisina arenosella Walker (Lepidoptera: Xylorictidae) in Karnataka

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ABSTRACT: Surveys in coastal and interior Karnataka during 1996 and 1997 revealed the occurrence of four larval, one larval-pupal and thirteen pupal parasitoids and eleven predators of Opisina arenosella Walker. Eight hyperparasitoids were also recorded.

KEY WORDS: Hyperparasitoids, Opisina arenosella, parasitoid, predator

The coconut leaf eating caterpillar, Opisina arenosella Walker is a serious defoliator of coconut, palmyra and wild date palm in India. It often assumes severe proportions in certain tracts. Coconut palms of all ages are infested by O. arenosella. Nimla et al. (1951) observed that massive damage to crown of leaves affected the thatching quality, in addition to decline in yield in the years following the outbreak of the pest. Perera (1993) est\mated the economic injury level at 15.5 per cent defoliation per palm and reported that the pest causing over 40 per cent visually assessed leaf damage can be considered as an outbreak.

Adverse effects of chemical pesticides administered by stem injection, root feeding and spraying have prompted the research workers to select biological control as a long term strategy to manage the insect pests in this perennial crop. Earlier reports revealed that D. arenosella supports 40 parasitoids and 22 predators (Pillai and Nair, 1993; Veeresh et al., 1995), and 26 species of spiders (Sathiammaet al., 1987). Among these natural enemies, occurrence of larval parasitoids, Apanteles taragamae Viereck, Goniozus nephantidis (Muesebeck); pupal parasitoid, Brachymeria spp. (Nadarajan and Channabasavanna,

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1980; Pushpalatha and Veeresh, 1995); in 70 per cent alcohol for determining the larval predator, Parena nigrolineata taxonomic status. (Chaudoir) (Gulagannavar, 1987) and egg predator Cardiastethus affinis Poppies RESULTS AND DISCUSSION (Srinivasa, 1996) have been studied in Karnataka. In order to strengthen the In coastal and interior Kamataka, four profile of natural enemies of O. arenosella larval, one larvo-pupal, thirteen pupal in Karnataka, a survey was conducted parasitoids and eleven predators were during 1996 and 1997 in coastal and recorded from O. arenosella. The natural interior areas, where the pest infestation enemies recorded are listed in Tables 1 and ranged from 50-90 per cent. 2.

MATERIALS AND METHODS In coastal Karnataka, three larval parasitoids viz., Apanteles taragamae, Observations on the occurrence of O. Goniozus nephantidis and Bracon arenosella and its natural enemies at brevicomis and ten pupal parasitoids viz., monthly interval were made at Kalmady Antrocephalus hakonensis, and Malpe (coastal) and Maddur (interior), A. maculipennis, Brachymeria In coastal area the coconut gardens selected nephantidis, B. nosatoi, B. atteviae, as monocrop were located on riverbanks. B. lasus, Trichospilus pupivora, Interior areas covered irrigated mixed Xanthopimpla punctata, Eupelmus sp. and cropping system involving coffee, guava, Anastatoidea brachartonae were recorded. sapota, banana, shoeflower, rose, jasmine, etc. At both locations, observations were Apanteles taragamae was recorded made on seedlings « 5 year - old), young throughout the year. Among the pupal palms (5 to 15 years) and tall palms (above parasitoids, A. hakonensis and 15 years) of coconut. A. maculipennis were the first to appear in February and continued their parasitising Observations on O. arenosella (egg, activity up to May. Similarly, T. pupivora larva, pupa) and its natural enemies were was recorded from February to November. recorded in the middle 40-60 per cent Eupelmus sp. was available in March-April, leaflets of 20 per cent leaves (80 leaflets A. brachartonae in April-May and per palm) (George et ai., 1982). Larvae and B. brevicornis in April, November and pupae of the pest were kept in separate December. Goniozus nephantidis was glass jars covered with muslin cloth for the available from May to November and emergence of parasitoids and B. nephantidis, B. nosatoi, B. atteviae and hyperparasitoids. Predators were separated B. lasus from June to September. by brushing out oflarval galleries and tested in the laboratory for predatory efficiency. In interior Karnataka, three larval Natural enemies collected were preserved parasitoids, one larvo-pupal and 10 pupal

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parasitoids were recorded. Apanteles Karnataka. Similarly, Mal/ada astur was
taragamae, M. hutsoni, B. nephantidis, recorded only in interior Karnataka.
B. nosatoi, B. atteviae and B. lasus were
recorded throughout the year. Tetrastichus Eight hyperparasitoids were recorded
howardi and T. pupivora were available from the primary parasitoids in the coconut
from February to November, ecosystem. Apanteles taragamae was
A. hakonensis, A. maculipennis and hyperparasitised by maximum seven
A. phaeospilus in April-May, Bracon sp. species while Goniozus nephantidis by six.
in April, June and August, G. nephantidis Aphanogmus manilae, Eupelmus
in April, August and December and albotibialis, E. braconidis, Pediobius
G. gibbosus in October - December. imbreus and Tetrastichus howardi were the
hyperparasitoids recorded on both
Xanthopimpla punctata, Eupelmus sp. A. taragamae and G. nephantidis in coastal
and A. brachartonae were recorded only Karnataka. In interior Karnataka, A.
in coastal Karnataka, and A. phaeospilus, manilae, E. albotibialis and E. braconidis
M. hutsoni, T. howardi and G. gibbosus were common on A. taragamae, and
only in interior Karnataka. Out of the eleven G. nephantidis. Meteoridea hutsoni was
predators recorded, 9 were available in re~orded from A. taragamae and
coastal Karnataka and 7 in interior P. imbreus from G. nephantidis. Bracon
Karnataka. Cardiastethus exiguus and sp. was hyperparasitised by A. manilae and
Phytoseiulus sp. were recorded throughout P. imbreus in interior and Brachymeria spp.
the year in coastal Karnataka and the by E. albotibialis in coastal Karnataka,
former in interior Karnataka. Parena respectively. The only tachinid recorded
nigrolineata, an d Cardiocondyla was hyperparasitised by Nesolynx dipterae
wroughtoni were recorded only in coastal in interior Karnataka.

Table 1. Hymenopteran parasitoids of O. arenosella

Name	Family	Stage	Period	Region
Apanteles taragamae Viereck	Braconidae	Larva	Throughout the year	Coastal and Interior
Bracon brevicomis Wesmael	Braconidae	Larva	April, November, December	Coastal
Bracon sp.	Braconidae	Larva	April, June August	Interior

Goniozus (=Parasierola) Bethylidae Larva November- Coastal
nephantidis (Muesebeck) May
April, Interior
August,
December

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Name	Family	Stage	Period	Region
Antrocephalus hakonensis (Ashmead)	Chalcididae	Pupa	February - May	Coastal
A. maeulipennis (Cameron)			April- May	Interior
A. hakonensis (Ashmead)				
A. maeulipennis (Cameron)				
A. phaeospilus Waterston				
Braehymeria nephantidis Gahan	Chalcididae	Pupa	September- June	Coastal
B. nosatoi Habu				
B. atteviae Joseph, Narendran & Joy			Throughout the year	Interior
B. lasus (Walker)				
Triehospilus pupivora Ferriere	Eulophidae	Pupa	November- February	Coastal Interior
Xanthopimpla punctata (Fabricius)	Ichneumonidae	Pupa	June - March	Coastal
Eupelmus sp.	Eupelmidae	Pupa	March - April	Coastal
Anastatoidea brachartonae Gahan	Eupelmidae	Pupa	April- May	Coastal
Meteoridea hutsoni (Nixon)	Braconidae	Larval-pupal	Throughout the year	Interior
Tetrastfehus howardi (Olliff)	Eulophidae	Pupa	November- February	Interior
Goryphus gibbosus Jonathan and Gupta	Ichneumonidae	Pupa	October - December	Interior

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Natural enemy complex ofO. arenosella

Table 2. Predators of o. arenosella

Name	Taxonomic status	Stage attacked	Period	Region
Cardiastethus e~iguus Poppius Heteroptera : Anthocoridae	Egg and first instar larva	Throughout the year	Coastal and Interior	
Buchananiella sodalis Buchanan-White Anthocoridae	Heteroptera: Egg and first instar larva	August- September October- November	Coastal Interior	
Parena nigrolineata (Chaudoir) Carabidae	Coleoptera: Larva August	January - July-January	Coastal Interior	
lauravia sp. Coccinellidae	Coleoptera; Egg July- December	March - June	Coastal Interior	
Phytoseiulus sp. Phytoseiidae	Acarina: Egg	Throughout the year	Coastal	
Monomorium floricala (Jerdon) Formicidae	Hymenoptera : Egg, larva and pupa	March-May September- November	Interior	
Monomorium spp. Formicidae	Hymenoptera: Egg, larva & pupa	June- September	Coastal	
Cardiocondyla wroughtoni Foret Formicidae	Hymenoptera: Egg, larva & pupa	June- September	Coastal	
Crematogaster sp. Fonnicidae	Hymenoptera: Egg, larva & and pupa	June- September November- December	Coastal Interior	
Tapinoma sp. Dotichoderinae, Fonnicidae	Hymenoptera: Egg, larva, pupa	Junc- September July- September	Coastal Interior	
'Wallada astur (Banks) Chrysopidae	Neuroptera: Egg and first instar larva	July- December	Interior	

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Table 3. Hyperparasitoids recorded on different parasitoids of O. arenosella

Name	Family	Name of primary parasitoid (host)
Coastal region		
Aphanogmus manilae (Ashmead)	Ceraphronidae	Apanteles taragamae Viereck
Goniozus nephantidis (Muesebeck)		
Eupelmus sp.	Eupelmidae	A. taragamae
G. nephantidis		
Eurytoma albotibialis Ashmead	Eurytomidae	A. taragamae
G. nephantidis		
Brachymeria spp.		
E. braconidis Ferriere	Eurytomidae	A. taragamae
G. nephantidis		
Pediobius imbreus (Walker)	Eulophidae	A. taragamae
G. nephantidis		
Tetrastichus howardi (Olliff)	Eulophidae	A. taragamae
G. nephantidis		
Interior region		
,		
A. manilae (Ashmead)	Ceraphronidae	A. taragamae
G. nephantidis		
Bracon sp.		
.		
E. albotibialis Ashmead	Eurytomidae	A. taragamae
G. nephantidis		
E. braconidis Ferriere	Eurytomidae	A. taragamae
G. llephantidis		
Meteoridea hutsoni (Nixon)	Braconidae	A. taragamae
P. imbreus (Walker)	Eulophidae	Bracon sp.
0. nephantidis		
Newlynx, dipterae Risbec	Eulophidae	Indet. Tachinid
;		

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